

AI & the Future of Semiconductor Work

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Namal Summer School
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10xEngineers.ai

US | Pakistan | Spain

Bilal Zafar

Co-Founder, 10xEngineers

Formally, **Principle Engineer** at **Qualcomm** (2011-18); Co-led an RTL2GDS design group with **60 engineers** (30 in NA and 30 in India);

Graduated with a **PhD** in Computer Engineering from the **University of Southern California**, Los Angeles, specializing in reconfigurable networks on chip; worked at Intel Microprocessor Research Lab



Qualcomm



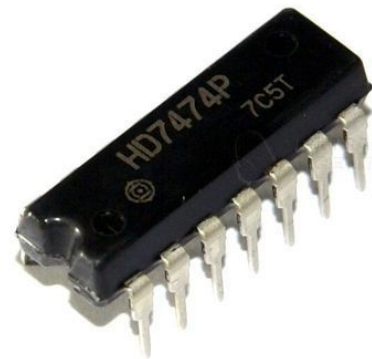
USC University of
Southern California



GIK INSTITUTE
OF ENGINEERING SCIENCES AND TECHNOLOGY

The Business of Making Chips

...





**POTATO
CHIPS**



**INTEGRATED
CIRCUITS**

Foundaries

Companies that
manufacture ICs



Fabless Semiconductor

Companies that
design ICs

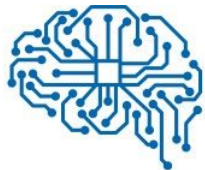


Services, Tools & IP

Companies that support
the design of ICs



Engineering Back Office



AQLTECH
SOLUTIONS

XCELERIUM



LOGICSILICON

GSME

Services Companies



ArcanaSemi

R
RENZYM

and many more...

Growing Semiconductor Industry

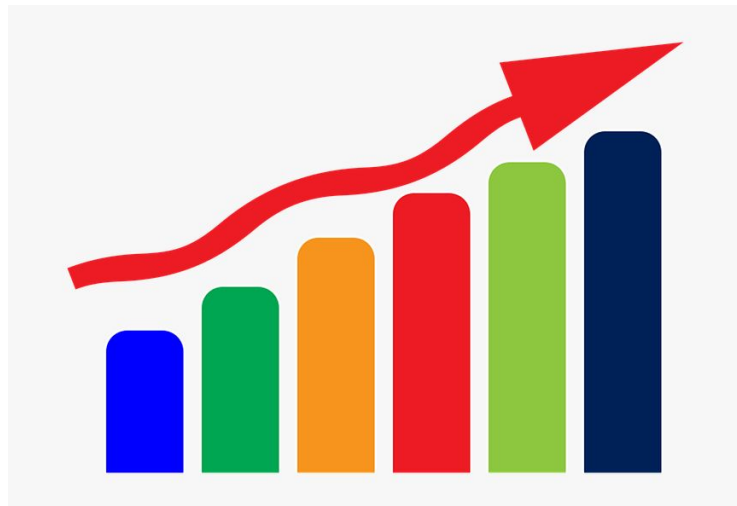
300K person shortfall the semiconductor industry worldwide

100+ jobs in Pakistan this year

Why Work in this Industry?

- Cutting-edge work
- Global employability
- Rapid growth

This is Engineering...



Jobs in the Industry

Circuit Design

Design SRAMs, analog
circuits

Physical Design

Place & Route

RTL Design

Specifications to
Verilog

Design Verification

Verification of RTL

Firmware Drivers

Bootloaders, Firmware,
Drivers, OS

Compilers Toolchain

Compilers, Libraries

10xEngineers Today

- Founded in 2021 as a hardware design services co
- Today: 120+ Engineers
- 9 In-House Training Cohorts; ~50 Engineers;
- 25+ Papers, Poster and Demos;
- Our Platforms
 - Cloud-V: RISC-V in the Cloud
 - Baltoro: AI Compiler for RISC-V Accelerators
 - QuantX: AI-Model Quantization
 - Infinite-ISP: An Open-Source
- Our IPs
 - RISC-V IOMMU, Video Scalar, HDR ISP



Four Areas



Design &
Verification

60%



Image
Processing

10%



AI Compilers

20%



Design Center
Hosting

10%

AI & Us

**“Do not confuse the headlines with
the trendlines”**

Bill Clinton

42nd President of the United States



Headlines

A.I. 'COULD WIPE OUT HUMANITY'

Threat 'as bad as nuclear war' MUST be tackled, say tech bosses



Trendlines

- People underestimate the rate at which AI models are improving

Training compute

↑ **5.2 months**

Training compute for frontier language models has been doubling every 5.2 months since 2020.

Compute stock growth

↑ **6.8 months**

The total computing power of the stock of AI chips is doubling every 6.8 months.

LLM context windows

↑ **2.4 months**

LLM context windows have been doubling every 2.4 months since 2023.



Trendlines

- People underestimate the rate at which AI models are improving
- **People underestimate the rate at which cost-per-token is decreasing**

Trendlines

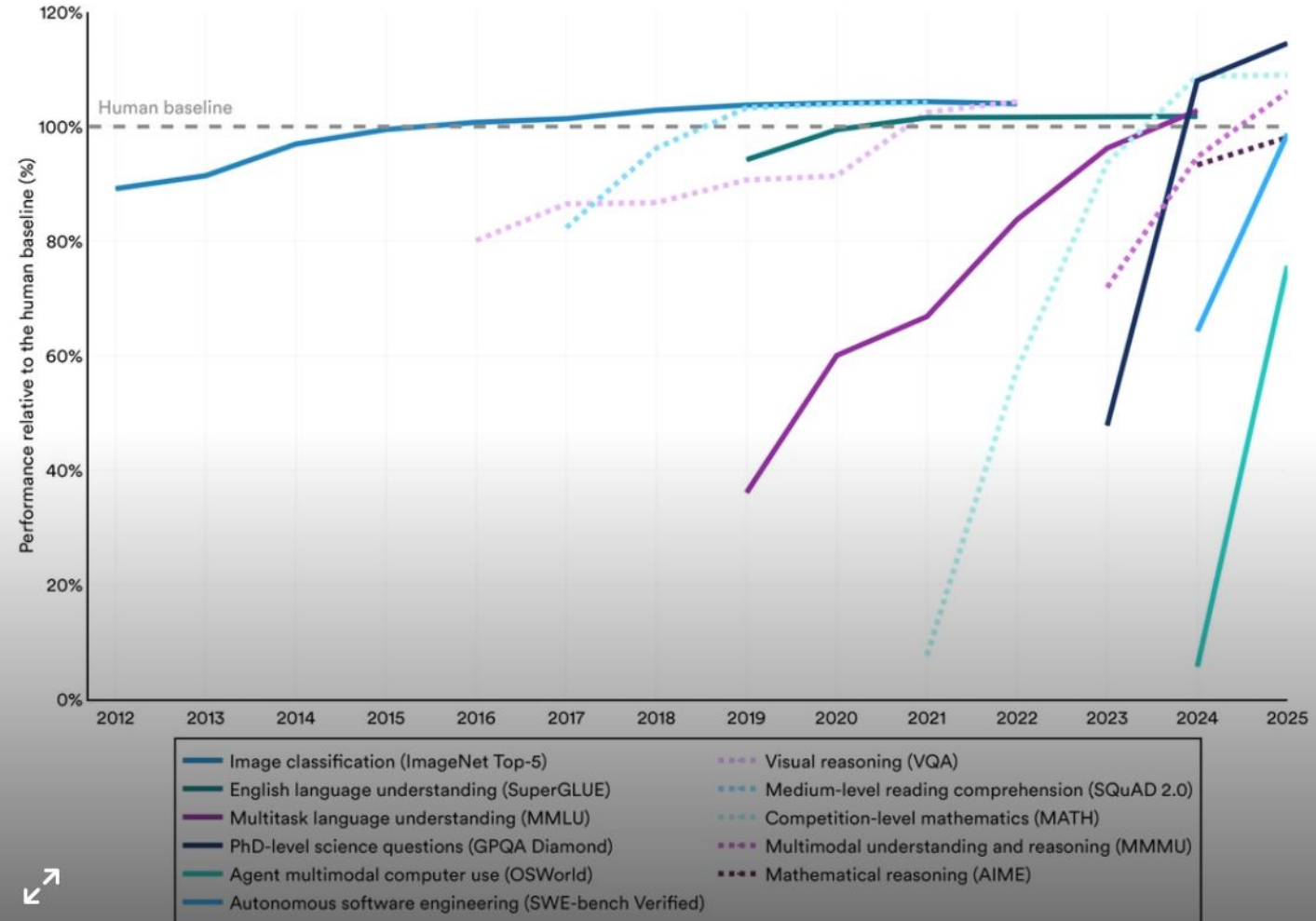
- People underestimate the rate at which AI models are improving
- People underestimate the rate at which cost-per-token is decreasing
- **People incorrectly expect the models to be smart in a “human” way**

Trendlines

- People underestimate the rate at which AI models are improving
- People underestimate the rate at which cost-per-token is decreasing
- People incorrectly expect the models to be smart in a “human” way
- **Students & teachers don't know how to prepare for the Age of AI**

Select AI Index technical performance benchmarks vs. human performance

Source: AI Index, 2026 | Chart: 2026 AI Index report



Source:
<https://hai.stanford.edu/ai-index/2025-ai-index-report/technical-performance>

What Should You Do?

STOP worrying about degrees, grades, GPAs.

No company or university that you SHOULD work for should care about these

ONLY focus on ONE thing:

How *fast* are you *learning* how to accomplish increasingly complex tasks?

Focus on what ***you can do***, not what others *should* do

Specific Step # 1: Learn to Solve Without and With AI

How to do Assignments or Projects?

1. Solve every assignment problem without AI first, then with AI
2. Conceive more complex versions of the problems, then try solving them without AI
3. Then solve them with the help of AI and understand what you did wrong
4. Complete every problem in the textbook (not just the few a teacher assigns)
5. Find equivalent course with more content and solve that

You can only focus on one sub-domain. EE or even CENG are too broad. Do it.

Specific Step # 2: Master the Tool

Master one AI Model ecosystem

Doesn't matter which one; just pick one and become a master of it
Anthropic, OpenAI, Gemini (or any other)

- Buy the subscription. This is the most important \$20/month you'll spend
- Learn every capability
- Play with it for hours for **one year** to master it at a level where you can be useful

I'm still figuring out more specific steps...

Final Advice: Focus

The hardest thing to do today is to **focus**

I am one of 100s of voices you'll hear on this topic in the next 30 days

You have to decide what to **focus** on